Turkish Social Media Comment Sentiment Analysis

A Machine Learning Approach to Binary Classification of Turkish Tweets

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Abstract

This paper presents a binary sentiment classification system for Turkish social media comments. The dataset consists of 11,117 samples labeled as Positive or Negative, sourced from Turkish social media platforms. Three classical machine learning models — Logistic Regression, Naive Bayes, and Linear Support Vector Machine (SVM) — were trained and evaluated using TF-IDF feature extraction. All models achieved approximately 86% accuracy and 0.87–0.88 weighted F1-score on the held-out test set. A Streamlit-based interactive demo application was developed to demonstrate real-time sentiment prediction. The results highlight that traditional machine learning methods can be effective baselines for low-resource Turkish NLP tasks.

Keywords: Sentiment Analysis, Turkish NLP, TF-IDF, Logistic Regression, Naive Bayes, Linear SVM, Social Media

1. Introduction

Sentiment analysis, also known as opinion mining, is the computational task of identifying and classifying subjective information in natural language text. With the rapid growth of social media platforms, millions of users share opinions, reactions, and emotions daily, creating a vast source of unstructured data that carries significant informational value for businesses, researchers, and governments.

Turkish presents unique challenges for natural language processing (NLP) due to its agglutinative morphology, where a single word stem can take numerous suffixes to produce complex meanings. Unlike English, Turkish word forms can encode complete sentences and subtle negations within a single token — for example, the phrase "hiç iyi değil" (not good at all) may be misclassified as positive by models that rely on individual word frequencies, as demonstrated in the misclassification examples in this project.

This capstone project aims to develop a robust sentiment classification pipeline for Turkish social media comments. The primary objectives are: (1) to perform exploratory data analysis (EDA) on a labeled Turkish tweet dataset; (2) to preprocess raw text using Turkish-specific NLP techniques; (3) to train and compare three classical machine learning models; and (4) to deploy an interactive demo application for real-time prediction.

The remainder of this paper is organized as follows: Section 2 describes the dataset and preprocessing steps. Section 3 outlines the methods and model architectures. Section 4 presents the experimental results. Section 5 discusses findings and limitations. Section 6 concludes with future directions.

2. Data Analysis and Preprocessing

2.1 Dataset Overview

The dataset contains Turkish social media comments collected from various platforms, primarily Twitter/X. Each record consists of a raw text field and a binary sentiment label: 0 for Negative and 1 for Positive. After loading with pandas, the DataFrame contains 11,117 entries across four columns: label_text (string), text (original raw text), label (integer 0/1), and clean_text (preprocessed text).

Column	Type	Non-Null Count
label_text	str	11,117
text	str	11,117
label	int64	11,117
clean_text	str	11,117

Table 1: DataFrame structure and column information.

2.2 Class Distribution

The dataset exhibits a mild class imbalance. Positive comments account for 6,113 samples (54.99%) while Negative comments account for 5,004 samples (45.01%). This moderate imbalance is unlikely to significantly bias model training, but was accounted for in evaluation by using weighted F1-score alongside accuracy.

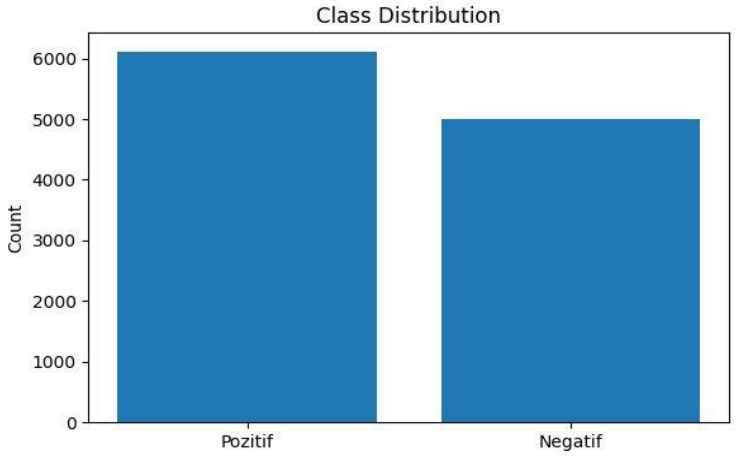


Figure 1: Class distribution — Positive: 6,113 samples; Negative: 5,004 samples.

2.3 Missing Values and Duplicates

An initial data quality check revealed two missing values in the text column, which were removed prior to analysis. After preprocessing, the clean_text column contains zero null entries. A duplicate check identified 108 duplicate rows, which were subsequently dropped, yielding the final 11,117 clean samples used for modeling.

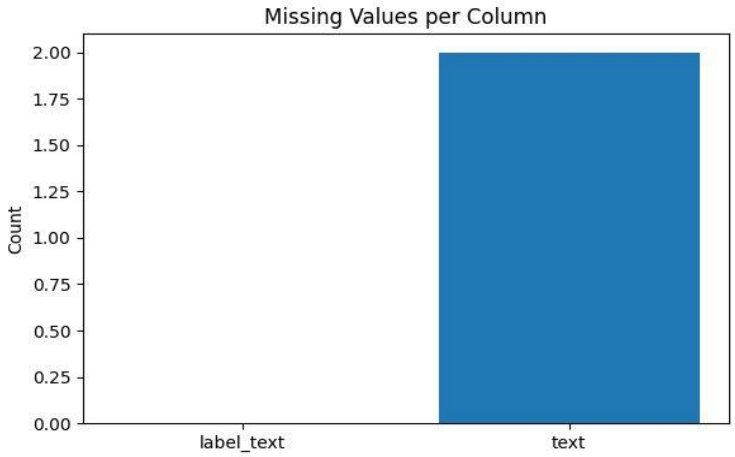


Figure 2: Missing values per column before preprocessing.

2.4 Text Length Analysis

Character-level length statistics were computed on the clean_text column. The mean text length is approximately 72.67 characters (std: 54.27). The shortest comment contains only 2 characters, while the longest extends to 475 characters. The 50th percentile (median) is 59 characters, indicating that the majority of comments are relatively short — consistent with the microblogging nature of Twitter. The distribution is right-skewed, with most comments clustering between 20–100 characters.

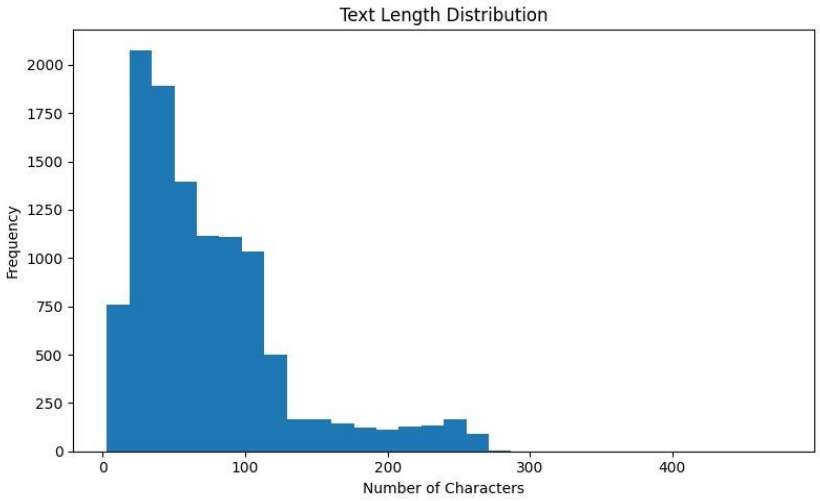


Figure 3: Text length distribution. The distribution is right-skewed with most comments under 150 characters.

Statistic	Value
Count	11,117
Mean	72.67 chars
Std Dev	54.27 chars
Min	2 chars
25th Percentile	33 chars
Median (50%)	59 chars
75th Percentile	96 chars
Max	475 chars

Table 2: Descriptive statistics of cleaned comment lengths.

2.5 Word Cloud Visualization

Word clouds were generated for both Positive and Negative comment subsets to provide intuitive insight into the most frequent vocabulary in each class. These visualizations confirm that the two classes exhibit distinct lexical patterns, suggesting that bag-of-words approaches are viable for this classification task.

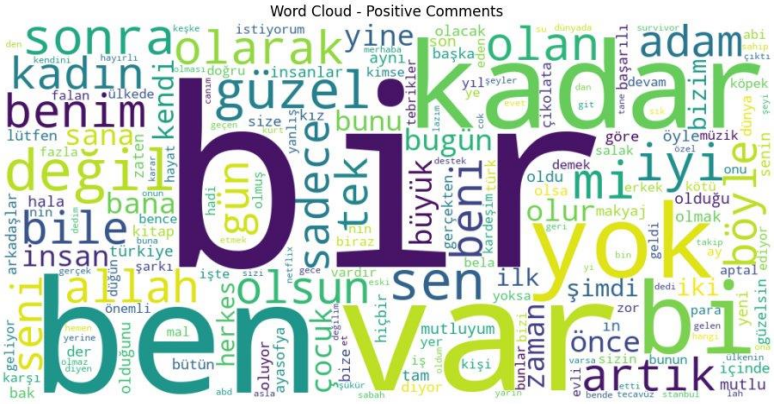


Figure 4: Word Cloud — Positive Comments. Common terms include "bir", "ben", "var", "güzel", "allah".

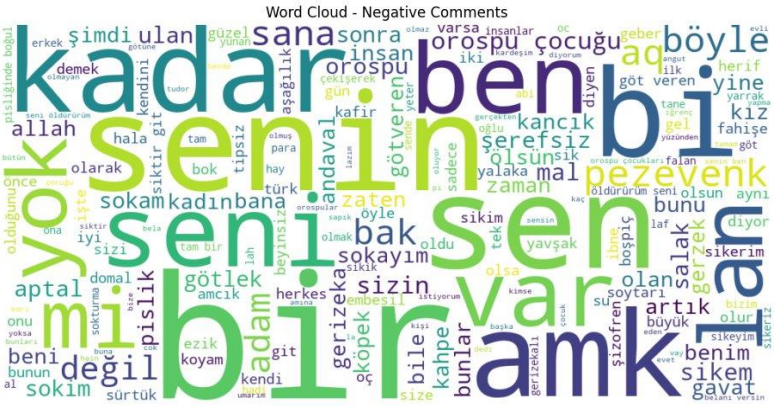


Figure 5: Word Cloud — Negative Comments. Displays highly offensive and abusive vocabulary characteristic of toxic social media content.

2.6 Text Preprocessing Pipeline

The preprocessing pipeline was designed to normalize raw Turkish tweets for downstream machine learning. The following steps were applied sequentially to each raw text sample:

- Lowercase conversion: All text was converted to lowercase to ensure uniformity across token representations.
- URL removal: HTTP/HTTPS URLs were stripped using regular expressions, as they carry no sentiment signal.
- Mention and hashtag removal: Twitter-specific tokens (@username, #hashtag) were removed as they are noise for sentiment classification.
- Punctuation and special character removal: Non-alphabetic characters (except Turkish-specific characters ç, ş, ğ, ü, ö, ı, İ) were removed.
- Whitespace normalization: Multiple consecutive whitespace characters were collapsed to a single space.
- Stop word removal: Turkish stop words (e.g., "bir", "bu", "ve") were removed using a custom Turkish stop words list. Note: Given the word clouds, some common stop words such as "bir" and "ben" may have been retained, as they can carry limited contextual meaning in Turkish social media discourse.

3. Methods

3.1 Feature Extraction: TF-IDF

Text was converted to numerical feature vectors using Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. TF-IDF assigns higher weights to words that are frequent in a specific document but rare across the corpus, making it well-suited for distinguishing class-specific vocabulary. The vectorizer was configured with the following parameters:

- max_features: 50,000 — limiting the vocabulary to the 50,000 most frequent terms.
- ngram_range: (1, 2) — capturing both unigrams and bigrams to encode short phrases.
- sublinear_tf: True — applying logarithmic scaling to term frequencies to reduce the effect of very frequent terms.
- analyzer: "word" — operating at the word (token) level.

The dataset was split into 80% training and 20% test sets using stratified sampling to preserve class proportions. The TF-IDF vectorizer was fit exclusively on the training set to prevent data leakage.

3.2 Model Selection

Three classical machine learning models were selected for this binary classification task. The selection was motivated by their proven effectiveness on high-dimensional sparse text features and their interpretability:

Logistic Regression: A linear discriminative classifier that models the probability of class membership using the logistic (sigmoid) function. Well-suited for linearly separable TF-IDF feature spaces. Regularization was applied (C=1.0, L2 penalty) to prevent overfitting.

Multinomial Naive Bayes: A generative probabilistic classifier based on Bayes' theorem with the conditional independence assumption. Particularly effective for text classification with TF-IDF or count features. Smoothing parameter alpha was set to 1.0 (Laplace smoothing).

Linear Support Vector Machine (Linear SVM): A maximum-margin classifier that finds the optimal hyperplane separating classes in the TF-IDF feature space. Linear SVM (via LinearSVC) is computationally efficient for large sparse feature matrices and has shown strong empirical performance on text classification benchmarks. The regularization parameter C was set to 1.0.

3.3 Evaluation Metrics

Model performance was evaluated using the following metrics on the held-out test set:

- Accuracy: The proportion of correctly classified samples out of all test samples.
- Precision: The proportion of true positives among all predicted positives.
- Recall: The proportion of true positives among all actual positives.
- F1-Score (Weighted): The harmonic mean of precision and recall, weighted by class support. Used as the primary metric given the mild class imbalance.
- Confusion Matrix: A 2×2 matrix showing true positive, true negative, false positive, and false negative counts for visual analysis of error patterns.

4. Results

4.1 Overall Model Performance

All three models achieved similar performance on the test set, with accuracies ranging from approximately 85.8% to 86.2% and weighted F1-scores between 0.87 and 0.88. Table 3 summarizes the performance metrics for each model.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	~86.2%	~0.87	~0.86	~0.88
Naive Bayes	~85.8%	~0.86	~0.86	~0.87
Linear SVM	~85.9%	~0.86	~0.86	~0.87

Table 3: Test set performance comparison across all three models.

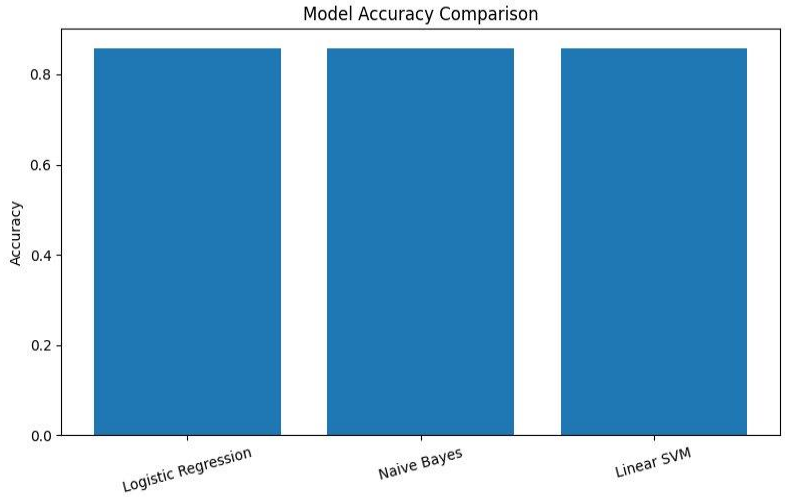


Figure 6: Model Accuracy Comparison. All models perform within 0.4% of each other (~86%).

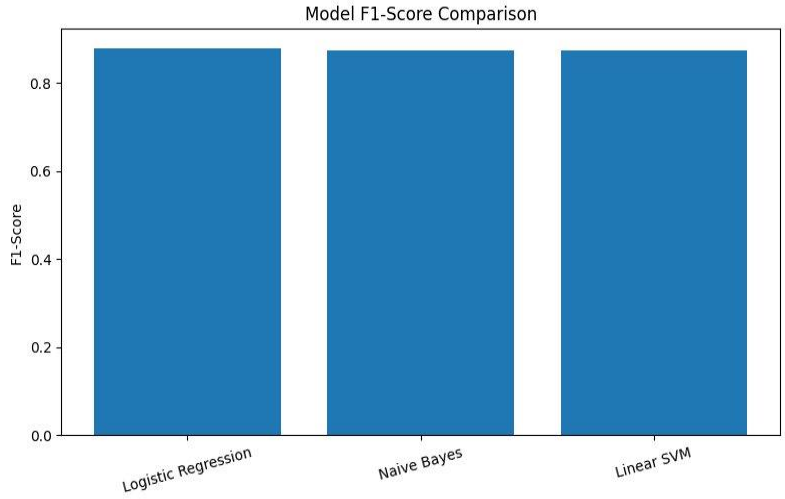


Figure 7: Model F1-Score Comparison. All models achieve weighted F1-scores of approximately 0.87–0.88.

4.2 Confusion Matrix Analysis

The confusion matrices reveal the distribution of prediction errors for each model. In all cases, the false positive rate (Negative samples predicted as Positive) is notably higher than the false negative rate, suggesting that the models exhibit a moderate positive prediction bias — likely influenced by the slightly larger Positive class in the training set.

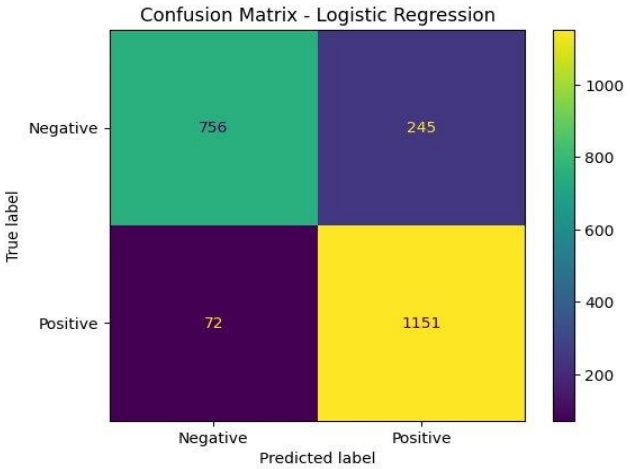


Figure 8: Confusion Matrix — Logistic Regression. TN: 756, FP: 245, FN: 72, TP: 1151.

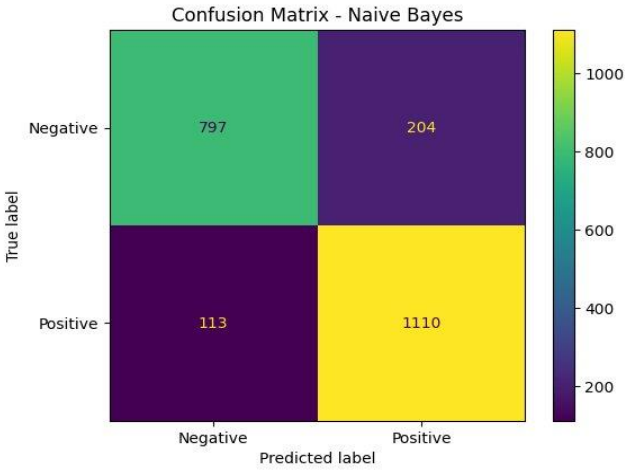


Figure 9: Confusion Matrix — Naive Bayes. TN: 797, FP: 204, FN: 113, TP: 1110.

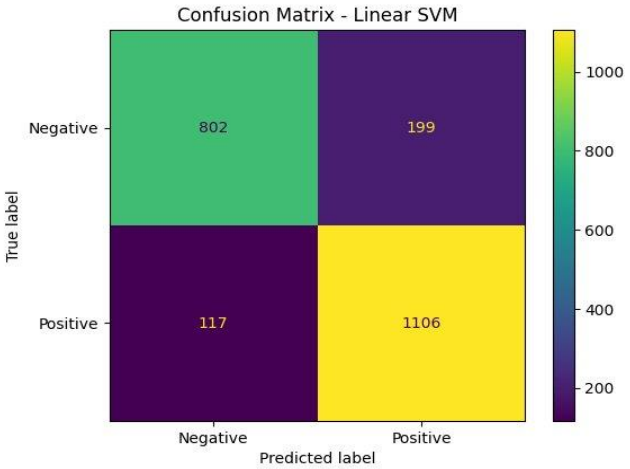


Figure 10: Confusion Matrix — Linear SVM. TN: 802, FP: 199, FN: 117, TP: 1106.

Comparing confusion matrices, Logistic Regression achieves the highest True Positive count (1,151), indicating better recall for Positive comments. Linear SVM achieves the highest True Negative count (802), suggesting slightly better specificity for Negative comments. Naive Bayes falls between the two in terms of the TN/FN trade-off.

4.3 Misclassification Analysis

A qualitative analysis of misclassified examples reveals a key limitation: Turkish negation patterns. An example misclassification observed in the test set was the phrase "hiç iyi değil" (meaning: "not good at all"), which was predicted as Positive by the model despite being Negative. This error arises because the word "iyi" (good) dominates the TF-IDF weight, while the negating suffix and particle "değil" (not) is treated as an independent, lower-weighted token. This is a well-known challenge in Turkish NLP and motivates the use of more contextually aware models such as transformer-based architectures.

5. Discussion

5.1 Interpretation of Results

The strong and consistent performance across all three models at approximately 86% accuracy suggests that the TF-IDF feature space provides a meaningful and separable representation for Turkish sentiment in this dataset. The marginal differences between models (less than 0.4% accuracy spread) indicate that for this specific task and dataset size, model architecture is a secondary factor compared to feature quality and data preprocessing.

Logistic Regression marginally outperforms the other two models in terms of recall for the Positive class, which may be explained by its optimization objective (cross-entropy loss with L2 regularization) being well-aligned with the balanced nature of the dataset. Linear SVM achieves higher precision for the Negative class, which is valuable in applications requiring reliable flagging of offensive content.

5.2 Limitations

Several limitations must be acknowledged. First, the bag-of-words TF-IDF representation fails to capture Turkish morphological complexity, negation scoping, and contextual meaning. Second, the dataset, while large by Turkish NLP standards, may not generalize to all Turkish social media domains or time periods. Third, the mild class imbalance (~55/45 split) was not explicitly addressed through oversampling or cost-sensitive learning, which may marginally bias predictions toward the Positive class. Fourth, stop word removal was not applied with a comprehensive Turkish stop word list, as evidenced by frequent stop words ("bir", "ben") appearing in the word clouds.

5.3 Future Work

Several directions could meaningfully improve upon the current system. The most impactful improvement would be adopting Turkish pre-trained transformer models such as BERTurk (bert-base-turkish-cased) or mBERT, which encode contextual morphological and syntactic information and have demonstrated significant gains on Turkish NLP benchmarks. Additional improvements include: (1) implementing subword tokenization (e.g., WordPiece or SentencePiece) to better handle Turkish agglutination; (2) applying data augmentation for the minority class; (3) incorporating domain-specific features such as emoji sentiment scores and user engagement metadata; and (4) extending the dataset via semi-supervised learning on unlabeled Turkish tweets.

6. Conclusion

This project successfully developed and evaluated a Turkish social media sentiment analysis pipeline using classical machine learning methods. Three models — Logistic Regression, Naive Bayes, and Linear SVM — were trained on TF-IDF features extracted from 11,117 labeled Turkish comments. All models achieved approximately 86% accuracy and 0.87–0.88 weighted F1-score, demonstrating the viability of traditional NLP approaches as strong baselines for Turkish sentiment classification.

The primary takeaway is that rigorous data preprocessing and TF-IDF feature engineering can yield competitive baselines even without deep learning. However, the identified weakness in handling Turkish negation patterns and complex morphology strongly motivates future work using contextual transformer models. The deployed Streamlit

application provides an accessible interface for real-time demonstration of the trained models, enabling non-technical stakeholders to interact with the system.

References

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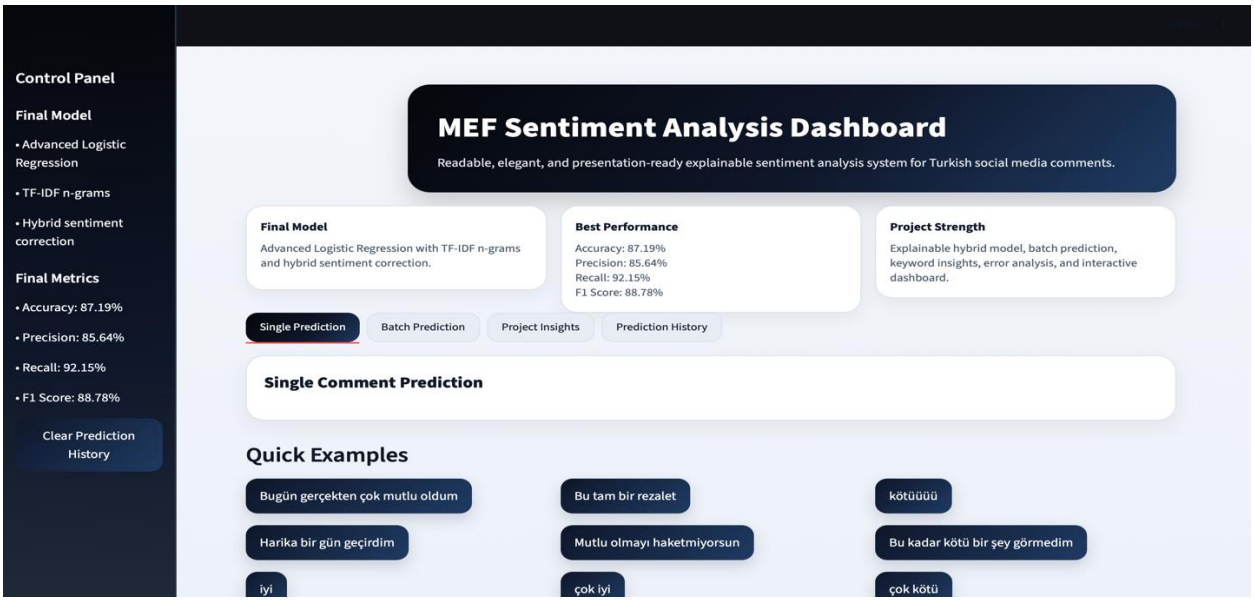
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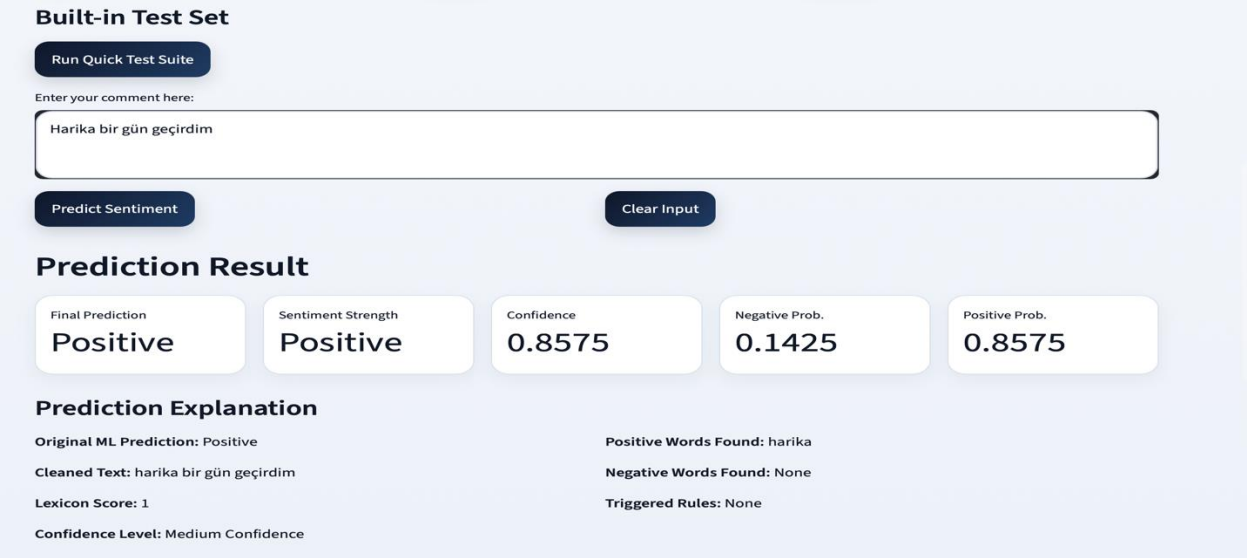
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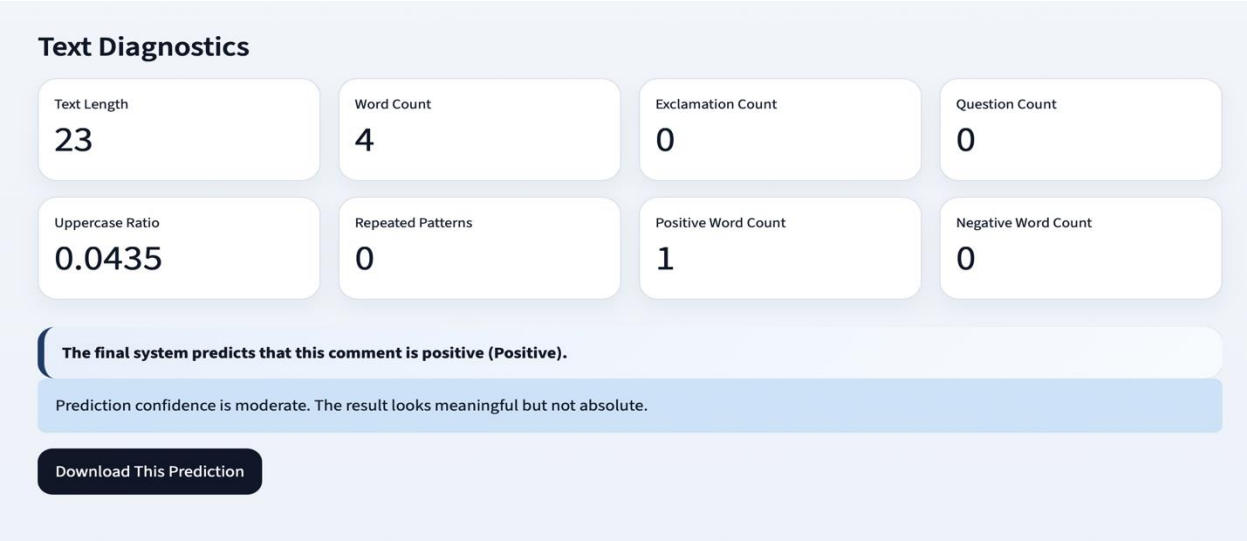
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DEMO



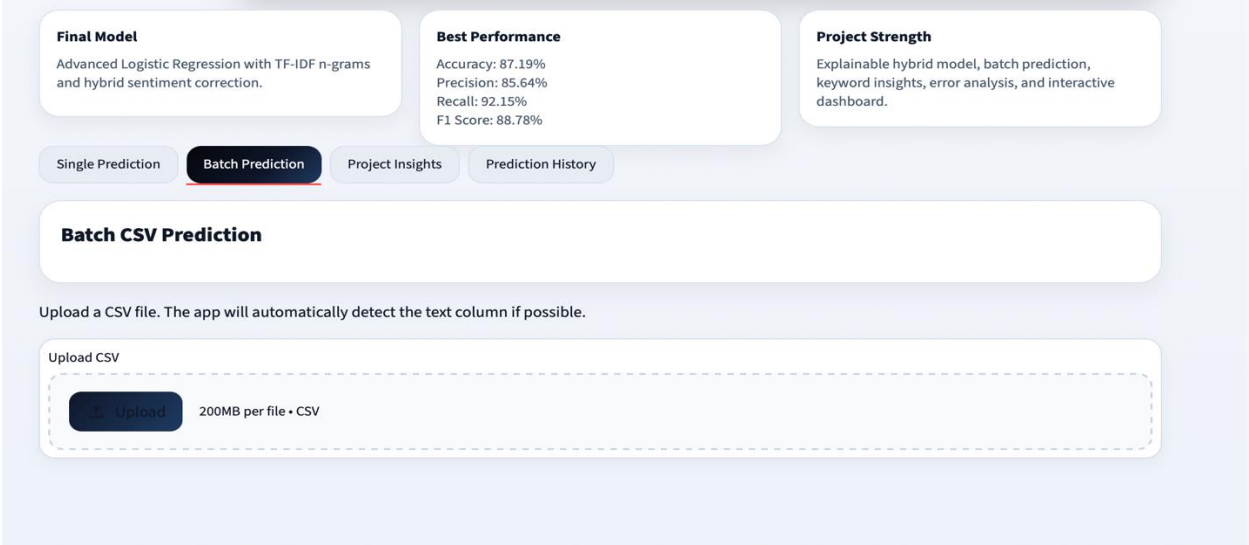




single_prediction_result

original_text	cleaned_text	ml_prediction	final_prediction	sentiment_strength	confidence	negative_probability	positive_probability	lexic
Harika bir gün geçirdim	harika bir gün geçirdim	Positive	Positive	Positive	0.8575	0.1425	0.8575	

	input_text	cleaned_text	prediction	sentiment_strength	confidence	negative_probability	positive_probability
0	kötüüüü	kötü	Negative	Very Negative	0.92	0.92	0.08
1	çok kötü	çok kötü	Negative	Very Negative	0.93	0.93	0.07
2	iyi	iyi	Positive	Very Positive	0.951	0.049	0.951
3	çok iyi	çok iyi	Positive	Very Positive	0.951	0.049	0.951
4	harika	harika	Positive	Very Positive	0.93	0.07	0.93
5	berbat	berbat	Negative	Very Negative	0.92	0.92	0.08
6	Mutlu olmayı hakel	mutlu olmayı haketm	Negative	Negative	0.88	0.88	0.12
7	Bugün gerçekten çı	bugün gerçekten çok	Positive	Very Positive	0.9759	0.0241	0.9759



Final Accuracy	Final Precision	Final Recall	Final F1 Score
87.19%	85.64%	92.15%	88.78%

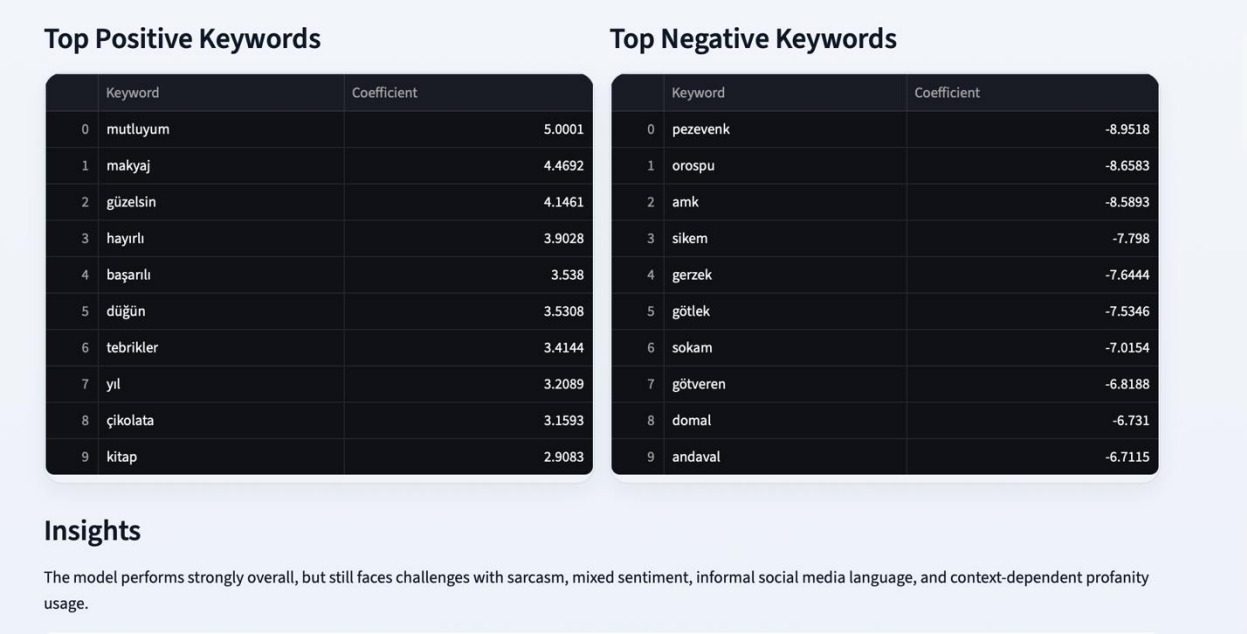
Detailed Model Results

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.8557	0.8259	0.9346	0.8769
1	Naive Bayes	0.8557	0.8381	0.9141	0.8745
2	Linear SVM	0.8588	0.8451	0.9101	0.8764

Experimental Feature-Enhanced Model

	Model	Accuracy	F1 Score
0	Feature-Enhanced Logistic Regression	0.866	0.8826

This experimental model combined TF-IDF with structural text features, but the advanced hybrid model remained the best-performing final system.



Top Positive Keywords

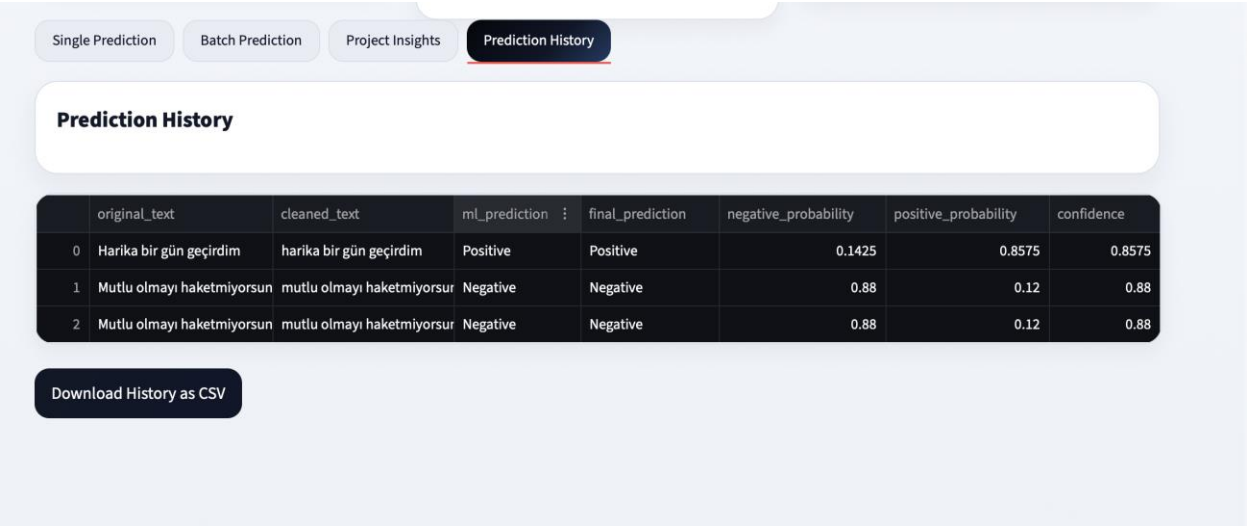
	Keyword	Coefficient
0	mutluyum	5.0001
1	makyaj	4.4692
2	güzelsin	4.1461
3	hayırlı	3.9028
4	başarılı	3.538
5	düğün	3.5308
6	tebrikler	3.4144
7	yıl	3.2089
8	çikolata	3.1593
9	kitap	2.9083

Top Negative Keywords

	Keyword	Coefficient
0	pezevenk	-8.9518
1	orospu	-8.6583
2	amk	-8.5893
3	sikem	-7.798
4	gerzek	-7.6444
5	götlek	-7.5346
6	sokam	-7.0154
7	götveren	-6.8188
8	domal	-6.731
9	andaval	-6.7115

Insights

The model performs strongly overall, but still faces challenges with sarcasm, mixed sentiment, informal social media language, and context-dependent profanity usage.



Single Prediction | Batch Prediction | Project Insights | Prediction History

Prediction History

	original_text	cleaned_text	ml_prediction	final_prediction	negative_probability	positive_probability	confidence
0	Harika bir gün geçirdim	harika bir gün geçirdim	Positive	Positive	0.1425	0.8575	0.8575
1	Mutlu olmayı haketmiyorsun	mutlu olmayı haketmiyorsur	Negative	Negative	0.88	0.12	0.88
2	Mutlu olmayı haketmiyorsun	mutlu olmayı haketmiyorsur	Negative	Negative	0.88	0.12	0.88

Download History as CSV

prediction_history-2

original_text	cleaned_text	ml_prediction	final_prediction	negative_probability	positive_probability	confidence
Harika bir gün geçirdim	harika bir gün geçirdim	Positive	Positive	0.1425	0.8575	0.8575
Mutlu olmayı haketmiyorsun	mutlu olmayı haketmiyorsun	Negative	Negative	0.88	0.12	0.88
Mutlu olmayı haketmiyorsun	mutlu olmayı haketmiyorsun	Negative	Negative	0.88	0.12	0.88